1. The Incremental Software Development Life Cycle Model

2. A. **Discussing the suitability of the chosen model to the problem you are trying to solve.**

   The Incremental SDLC Model offers numerous benefits for developing a healthcare system with multiple features. It enables iterative development and delivery of functionalities like a medical chatbot, virtual consultations, and health recommendations. This approach allows for faster feedback and validation, ensuring effective fulfillment of user requirements. The model promotes continuous improvement by incorporating user feedback, enhancing features, and improving the overall user experience over time. It offers flexibility and adaptability to address changing requirements, emerging technologies, and evolving user needs. By mitigating risks through incremental testing and validation, it ensures patient safety and data security. Additionally, it enables a quicker time-to-market by prioritizing essential features and gaining early user feedback. Ultimately, the Incremental SDLC Model embraces a user-centric approach by involving users and stakeholders throughout the development process, resulting in a healthcare system that meets their specific needs and delivers optimal user satisfaction.

   B. **State and discuss how this model will support the background concepts related to the problem.**

   The Incremental SDLC Model is user-centered and involves users throughout the development process. It is flexible and can adapt to changes in healthcare. It continuously improves by gathering feedback and making enhancements in each stage. It also reduces risks by testing and addressing issues early. The model delivers functional parts quickly, providing value to users and staying ahead of competitors. Overall, it focuses on users, adapts to changes, improves continuously, manages risks, and delivers quickly in healthcare system development.

Risk mitigation refers to the process of identifying, assessing, and implementing measures to reduce or eliminate potential risks and their negative impacts on a project or system. It involves proactively identifying and analyzing risks, understanding their likelihood and potential consequences, and developing strategies to minimize their occurrence or impact. Risk mitigation aims to protect project objectives, resources, and stakeholders by implementing preventive actions, contingency plans, and risk response strategies. By mitigating risks, organizations can enhance project success, improve decision-making, increase stakeholder confidence, and ensure the overall safety and security of the system or project being developed.